

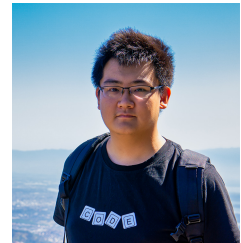
# Mo Sha

## Ph.D. Candidate in Computer Science

@ sham@comp.nus.edu.sg

🌐 www.comp.nus.edu.sg/~sham/

✉ COM1-108, 13 Computing Drive, Singapore 117417



## SELECTED PUBLICATIONS

- **Self-adaptive Graph Traversal on GPUs.**  
M Sha, Y Li, and KL Tan.  
International Conference on Management of Data (**SIGMOD**), 2021.
- **GPU-Accelerated Subgraph Enumeration on Partitioned Graphs.**  
W Guo, Y Li, M Sha, B He, X Xiao, and KL Tan.  
International Conference on Management of Data (**SIGMOD**), 2020.
- **GPU-based Graph Traversal on Compressed Graphs.**  
M Sha, Y Li, and KL Tan.  
International Conference on Management of Data (**SIGMOD**), 2019.
- **Accelerating Dynamic Graph Analytics on GPUs.**  
M Sha, Y Li, B He, and KL Tan.  
International Conference on Very Large Data Bases (**VLDB**), 2018.
- **Parallel Personalized Pagerank on Dynamic Graphs.**  
W Guo, Y Li, M Sha, and KL Tan.  
International Conference on Very Large Data Bases (**VLDB**), 2018.

## PATENT

- **Cross-section extraction for vortex detection.**  
M Sha, J Liu, Y Ke, E Leonardi, and K Menzies.  
Rolls-Royce, US20190303515A1/EP3547174A1, filed on Mar 2019.

## HONORS & AWARDS

- Dean's Graduate Research Excellence Award, NUS, 2020.
- Research Achievement Award, NUS, 2018.
- Gold Medal, ACM/ICPC Asia Regional, ACM, 2012.
- The CCF Outstanding Undergraduate Award, CCF, 2012.

## PROFESSIONAL SERVICES

### External Reviewer:

- International Conference on Very Large Data Bases, 2021.
- IEEE International Conference on Data Engineering, 2019.

### Invited Reviewer:

- Transactions on Knowledge and Data Engineering, 2021.
- Journal of Parallel and Distributed Computing, 2019.
- IEEE Access, 2019.

## TEACHING

### Teaching Assistant:

- CS5234: Algorithm at Scale, NUS, 2019.
- CS3223: Database Systems Implementation, NUS, 2018-2021.
- CS1101S: Programming Methodology, NUS, 2018.

## EDUCATION

### Ph.D. in Computer Science

supervised by: [Prof. Kian-Lee Tan](#)

**National University of Singapore**

📅 Aug 2017 – 2021(expected) 📍 Singapore

### B.Eng. in Computer Science

**Harbin Institute of Technology**

📅 Sep 2010 – Jun 2014 📍 Harbin, China

## EXPERIENCES

### Research Assistant

**Database Lab, School of Computing, NUS**

📅 Jun 2016 – Jun 2017 📍 Singapore

Our team was interested in dynamic graph analytics on various applications and was exploring to utilize different hardware hierarchies, e.g., GPU accelerators and their combined architecture, for higher efficiency.

### Project Officer

**Rolls-Royce@NTU Corporate Lab, NTU**

📅 Jun 2015 – Jun 2016 📍 Singapore

I took part in the research project focusing on the automatic vortex extraction and tracking in CFD data, which is an essential tool for the aerodynamic design of Rolls-Royce's jet engine.

### Algorithm Engineer

**Alibaba**

📅 Jun 2014 – Jun 2015 📍 Beijing, China

I was responsible for the backend of the business logic layer of internet advertisement delivery engine, which was mainly related to user profiling and targeted advertisement indexing, retrieving and recalling.

### Software Engineering Intern

**Google**

📅 Jun 2013 – Nov 2013 📍 Shanghai, China

Our team was in charge of developing the backend transaction logic management of the Demand Side Platform, and my main contribution during the internship was the RetryAction module.